Waze User Churn Analysis | Preliminary Data Summary

Prepared for: Waze Leadership Team

OVERVIEW

This project analyzes user churn and driving behavior using an educational Waze dataset containing 14,999 records. The analysis examines data quality, missing values, and differences between churned and retained users to identify patterns that may inform future retention strategies.

This report summarizes the key findings from the preliminary analysis of Waze user churn, highlighting data quality, driving behavior, and potential factors associated with user retention.

PROJECT STATUS

Milestone 2 – Preliminary Data Analysis Completed

🎯 **Target Goal:** Inspect Waze user data to assess data quality, identify missing values, and compare driving behavior between churned and retained users.

🎯 **Methods:**

- Loaded and inspected 14,999 user records using Python and Pandas.
- Identified 700 missing churn labels.
- Calculated descriptive statistics and compared churned versus retained users.
- Analyzed driving patterns and device distributions.

🎯 **Impact:** The analysis identified meaningful differences in driving behavior between churned and retained users. These findings provide a foundation for further investigation into user churn and potential retention strategies.

NEXT STEPS

➔ **Investigate high-activity users:** Gather additional information about users who drive long distances and make frequent trips to understand their navigation needs and possible reasons for churn.

➔ **Conduct further analysis:** Investigate the relationship between driving activity and user churn, examine the 700 missing churn labels, and explore additional factors such as user satisfaction and app performance before developing a predictive model.

KEY INSIGHTS

- **Missing Data:** The dataset contains 14,999 user records, with 700 missing churn labels (4.67%). The remaining 14,299 records include 11,763 retained users and 2,536 churned users.
- **Driving Behavior:** Churned users had a higher median number of drives per driving day (10.0) than retained users (4.06). They also drove more kilometers per driving day (697.54 km vs. 289.55 km).
- **Device Comparison:** Churn rates were nearly identical across devices, with approximately 17.83% of iPhone users and 17.54% of Android users churning. Device type does not appear to be strongly associated with churn in this dataset.
- **Distance per Drive:** The median distance per drive was similar for churned users (74.11 km) and retained users (75.01 km). This suggests that driving frequency and daily distance may be more informative for further churn analysis than distance per individual drive.
- **User Retention:** Among the 14,299 users with known churn labels, approximately 82.26% were retained and 17.74% churned. This provides a baseline for future churn analysis and predictive modeling.